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RNA-Interference-Mediated Silencing of OCT4B1, Alters Expression Profile of Several TNF Ligand/ Receptor Transcripts in Human Tumor Cell Lines

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ABSTRACT

Background: The OCT4B1 as a new discovered variant of OCT4 is expressed in both cancer cell and tissues. This to scape from apoptosis. TNF ligands and receptors are amongst two categories of eleven gene families involved in the apoptosis pathway. Therefore, the aim of the present study was to investigate the effects of OCT4B1 suppression on ***Methods:***The AGS (gastric adenocarcinoma), 5637 (bladder tumor) and U-87MG (brain tumor) tumor cell lines were transfected with specific OCT4B1 siRNA and , as well as a scrambled sequence and parallel with beta-actin (as housekeeping gene) using Real-Time PCR technique. ***Results:*** our results indicated that in TNF ligand transcripts family, the mRNA level of TNF transcripts was up-regulated and inversely TNFSF8, TNFSF7, TNFSF10, TNFSF1 and TNFSF6 was down-regulated. We observed also that in TNF receptor transcripts family, six transcripts including, TNFRSF 10A, TNFRSF10B, TNFRSF11B, TNFRSF1A, TNFRSF21 and TNFRSF25 were up-regulated, while TNFRSF9 and CD27 were down-regulated. ***Conclusions:*** According to these results, it may be concluded that OCT4B1 suppression can lead to apoptosis in tumor cell lines via up-regulation of several TNF ligand and receptor transcripts. Thus, OCT4B1 suppression effects on TNF and its receptors may be considered as promising target genes in future studies in cancr research and therapy.

Introduction

OCT4 genes (octamer DNA binding transcription factor 4), belong to a family of transcription regulatory genes containing the PUO DNA binding domain, located on human chromosome 6 that potentially encodes at least three different variants (A, B and B1) via mRNA alternative splicing.[1]OCT4 is among the important and critical genes involved in both embryonic and adult stem cell proliferation and differentiation.[2] Expression of OCT4 induces stem cell pluripotency, while, it's silencing or down regulation is believed to be a sign of cell differentiation.[2] According to the most recent theories of cancer known as "cancer stem cell", cancers are originated from either adult stem cells (tissue stem cells) or reprogrammed tissue somatic cells.[3] There are gene expression pattern similarities between cancer stem cells and normal stem cells.[4] Recently it has been established that a new variant of OCT4, known as OCT4B1, is expressed in both normal stem cells (embryonic and adult) and cancer cell

lines/cancer tissues[1]. Moreover, investigations showed that this variant has anti-apoptotic potency in cancer cell lines.[5] Apoptosis, an important cellular pathway, occurs continually during development and aging to preserves cell populations in tissues.[6] There are at least twelve gene families related to apoptotic pathway, of which TNF ligand and TNF receptor genes are subclasses of these families.[7]TNF ligands and receptors not only play important roles in apoptosis but are also involved in normal developmental processes including regulation of immune cell functions and additional cell type-specific responses.[8]

TNF ligands are a set of genes that encode cytokines polypeptides with the ability to activate TNF receptor proteins located in the surface of cells.[9] This superfamily consists of 19 ligands recognized as TNFSF (tumor necrosis factor superfamily), most of which are type II of transmembrane proteins whose

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extracellular domains can be cleaved by specific metalloprotease to generate soluble cytokines[10,11]. TNF receptors are a set of 29 genes called TNFRSF (tumor necrosis factor receptor) oligomeric, and belong to type I or type III transmembrane proteins. Several TNF receptors contain death domains (DDs) that recruit caspase-interacting proteins following ligand binding to initiate the extrinsic pathway of caspases activation.[9] Additionally, due to the anti-apoptotic properties of OCT4B1, one hypothesis suggests that it can affect the expression of essential genes involved in cell survival and cell death, especially programmed cell death (apoptosis) to induce the self-renewal state of the cell.

Apoptosis is a specialized mechanism for removal of the aggressive cells including cancer cells; however, within the cancer tissues and cancer cell lines, apoptosis would be suppressed. Delineation of both cellular and molecular mechanisms that contribute to these suppression processes could be considered as promising targets in cancer therapy.

Based on the aforementioned introductory facts, because OCT4B1 can inhibit apoptosis, the aim of the present investigation was to explore the relation between the OCT4B1 suppression and expression profile of several TNF ligand and TNF receptor family

genes in three human tumor cell lines including AGS (gastric adenocarcinoma), 5637 (bladder tumour) and U-87MG (brain tumour).

Materials and Methods

Cell culture and transfection

AGS (gastric adenocarcinoma), 5637 (bladder tumour) and U-87MG (brain tumour) were maintained in RPMI-1640 (Gibco) culture medium containing 10% fetal bovine serum (Gibco) and antibiotics (100 mg/mL streptomycin and 100 U/mL penicillin, Gibco company) at 37 °C in a humidified incubator containing 5% CO2. The cells were subcultured three times (to assurance of cell viability) and then cultured in tow test and control groups (10_5 cells/ml). To suppress OCT4B1 variant, two specific siRNA were utilized based on the specific OCT4B1 sequences. One irrelevant or scrambled siRNA (with no complementary target sequence in the human genome) and a vehicle in PBS mock were transfected under the same conditions as controls. The siRNA were designed by a selection program (Whitehead Institute for Biomedical Research, htt://jura.wi.mit.edu/) and the process of synthesis was performed by MWG Company (Germany) (Table 1).

Table 1. sequences and criteria of designed siRNAs.

siRNA name		**Target**	**Sequences**
Version I		Target	AAGGAGTA
		Sense	(GGAGUAU
		Anti-sense	(CUAGGUU
Version II		Target	AAGAGGTG
		Sense	(CAGUGGU UC)dTdT
Scramble	Sense	GCGGAGAGGCUUAGGUGUA	

Anti-sense	UACACCUAAGCCUCUCCGCdTdT

For siRNA transfection, lipofection method was used; briefly, a suspension of freshly cultured cells was prepared (10^5 cells/ml) in the RPMI1640 culture media without antibiotics, seeded in six well plates in two test and control groups. During logarithmic growth phase (confluences 30-50%), cells were transfected with 50 nmol/ml synthetic siRNA (in the control group, scrambled sequence siRNA), using lipofectamin 2000 (Invitrogen, USA) and Opti-MEM media, according to the manufacturers' instructions. In brief, 5µl of siRNA (25µM) and 4.5 µl RNAi-MAX reagents were diluted in 250 µl Opti-MEM and incubated for 10 min at room temperature. The mixture was then added to the cells in a final volume of 2.5ml, and then the cells were incubated at 37ºC in a humid atmosphere of 5% CO_2

cell lines, total RNA was extracted from cultured cells (10^6 cells/ml) using TRIzol reagent (Invitrogen, UK). The contaminated DNA in the RNA samples was

removed by TURBO DNase I (Fermentase, Lithuania). The RNA fidelity purity and integrity were measured by spectrophotometer (260/280 NM ratio) and gel electrophoresis (1% agarose gel). The first strand cDNA was synthesized at 42ºC for 60 min using 100 pmol oligo-dT primer, 200 units of MMuLV reverse transcriptase and 1µg of total RNA according to the manufacturer's instructions (Fermentase, Lithuania). Quantitative Real Time PCR (polymerase chain reaction) was performed to detect OCT4 variants gene expression in studied cells as described later. Specific primers were designed for OCT4 variants (A, B and B1), interested TNF ligands and receptors transcripts and ß-actin, using Gene Runner (version 3.02) and for 72 hours.

The OCT4 Allele ID (version 4.0) software (Table 2). OCT4 variant

Table2. Demonstrates designed primers for OCT4 variants, TNF ligand and receptor genes families and ß-actin.

Target genes (Gene Symbol)	Gene bank number	Designed Oligo	Relative Sequence	Fragment length
OCT4A	NM-002701	F	CGCAAGCCCTCAT	**111**
		R	CATCACCTCCACCACCTG	
OCT4B	NM-203289	F	CAGGGAATGGGTG	**177**
		R	AGGCAGAAGACTTGTAAGAAC	
OCT4B1	EU518650	F	GGTTCTATTTGGTG	**128**
		R	TTCTCCCTCTCCCTACTCCTC	
CD40LG	NM-000074	F	AACCCTGGAAAAT	**222**
		R	CCTCCCAAGTGAATGGATTG	
FASLG	NM-000639	F	TGGCCTTGTGATC	**155**
		R	TCATCATCTTCCCCTCCATC	
LTA	NM-000595	F	CCCACCAGTGGCA	**174**
		R	CAGCCCTGGATACACCATCT	
TNF	NM-000594	F	CAGAGGGCCTGTA	**219**
		R	GGAAGACCCCTCCCAGATAG	
TNFSF10	NM-003810	F	TTCACAGTGCTCCT	**192**
		R	ACGGAGTTGCCACTTGACTT	
CD70	NM-001252	F	TGGTACACATCCA	**212**
		R	AAGTGTCCCAGTGAGGTTGG	
TNFSF8	NM-001244	F	ACTCTGGCTCTGT	**152**
		R	AATGGAGCCCTTTTCAGGAT	
CD40	NM-001250	F	GCAGGCACAAACA	**191**
		R	TCGGGAAAATTGATCTCCTG	
FAS	NM-000043	F	CAAGGGATTGGAA	**203**
		R	TGGAAGAAAAATGGGCTTTG	
LTBR	NM-002342	F	ACCAGGTGTGAGA	**153**
		R	GAGCAGAAAGAAGGCCAGTG	
TNFRSF10A	NM-003844	F	AGAGAGAAGTCCC	**154**
		R	GTCACTCCAGGGCGTACAAT	
TNFRSF10B	NM-003842	F	CACCAGGTGTGAT	**221**
		R	CCCCACTGTGCTTTGTACCT	
TNFRSF11B	NM-002546	F	GGCAACACAGCTC	**241**
		R	CTGGGTTTGCATGCCTTTAT	
TNFRSF1A	NM-001065	F	GTGCCTACCCCAG	**175**
		R	TGTCGATTTCCCACAAACAA	
TNFRSF21	NM-014452	F	TGAACAAGACCCT	**215**
		R	CAGCAGGAAAAGCACAATCA	
TNFRSF25	NM-003790	F	GTGTGTCCCCAAG	**206**
		R	GGTTGGCAGTAGAAGGGTGA	
CD27	NM-001242	F	CAGCCCACCCACT	**226**
		R	TCCTTCGTTGATGGAGGAAC	
TNFRSF9	NM-001561	F	CACTCTGTTGCTG	**158**
		R	CACAGGTCCTTTGTCCACCT	
ß-actin	NM-001101	F	AGGCACCAGGGCG	**184**
		R	GCCCACATAGGAATCCTTCTGAC	

MTT Assay

The cell viability was determined using 3-(4, 5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay on the basis of the ability of viable cells to metabolize MTT. MTT is a water-soluble tetrazolium salt which can be metabolized into a water-insoluble formazan product by mitochondrial succinate dehydrogenase in viable cells.[12] Briefly, 10^4

were seeded onto a 96-well plate at a final volume of 200 μL. Subsequent to cell transfection, the resultant supernatant was replaced by 200 μL of pre-warmed
RPMI 1640 (without phenol red) and continued by addition of 10 μL of 5 mg/ml MTT to each well. The plate was incubated at 37 ˚C for 3.5 h in the dark until a

purple precipitate was visible under light microscopy. Finally, 200 μL of DMSO (dimethyl sulphoxidase) was added to each well and 15 min after the absorbance (at 570 nm) was read after with a reference filter of 620

USA), 200ng of the generated cDNA and 2 pgr/µl of appropriate primers (Table 2). The following cycling program was set on a BIO-RAD CFX96 system (Bio-Rad Company, USA): one cycle at 95ºC for 15 min, 40 cycles at 95ºC for 30 S, 58-61ºC for 30 S and 72ºC for 30 S, 58-61ºC for 30 S and 72ºC for nm.

Apoptosis analysis

The ratio of apoptotic cell in OCT4B1 siRNA transfected cells (test group) against the scrambled sequence siRNA transfected cell (control group) was detected 48 hours after transfection. Single cell suspension was carried out, and the Annexin V- FLOUS kit (Roch, Germany) was used for detection of the exposed phosphatidylserin on cell surfaces, as described elsewhere.[13] Then, 48 hours after transfection, cells (1×10^6) were centrifuged and resuspended in 200µl of binding buffer. Annexin V-FLOUS (1µl) and propidium iodide (1µl) were added to tubes after 5 minutes and were further incubated for 5 minutes at room temperature (in the dark) and then

30 S. Real-time PCR was carried out in triplicate and ß-actin was assessed as a housekeeping gene for the normalization of amplification signals of the target genes. The proportional amounts of PCR products were determined using the $2^{-\Delta\Delta Ct}$ formula. Relative expression is determined by comparison of interested gene expression in test and control group using data from the real-time cycler and the ΔΔCT method. The dissociation stages, melting curves and quantitative analyses of the data were performed using CFX manager software version 1.1.308.111 (Bio-Rad, USA). All PCR products were visualized by electrophoresis on a 1% gel containing 0.5 mg/ml ethidium bromide to check the size of PCR products.

analyzed by Beckman-Coulter Elite flow cytometer. Annexin V-FLOUS binding was detected using a FITC detector (FL1) and propidium iodide staining by the phycoerythrin emission signal detector (FL3). The ratio of FITC/PI positive cells (apoptosis) was calculated from flow histograms.[14]

TNF ligand and receptor transcripts profiling

Followed suppression of OCT4B1 variant, mRNA synthesis of seven transcriptsin TNF ligand and eleven

Results

siRNA transfection caused down-regulation of signal *OCT4B1*

transcriptsin TNF receptor gene families were detected. Total RNA from test and control group were extracted and cDNA was synthesized as described. **BM quantum computers leverage furniture Real Time PCR was performed by the** addition of a SYBER green master mix (Qiagene,

The expression of OCT4B1 variant was assessed at 24, 48 and 72 hour time points, following siRNA transfection in test (OCT4B1 siRNA transfected cells) and control (scrambled sequence siRNA transfected cells) groups. In the test group version I and II of designing siRNA was carried out in triplicate and the results showed more suppression in the version I siRNA transfection. Our data showed that optimal knockdown of the gene occurred after 48 hours of treatment (Figure 1).

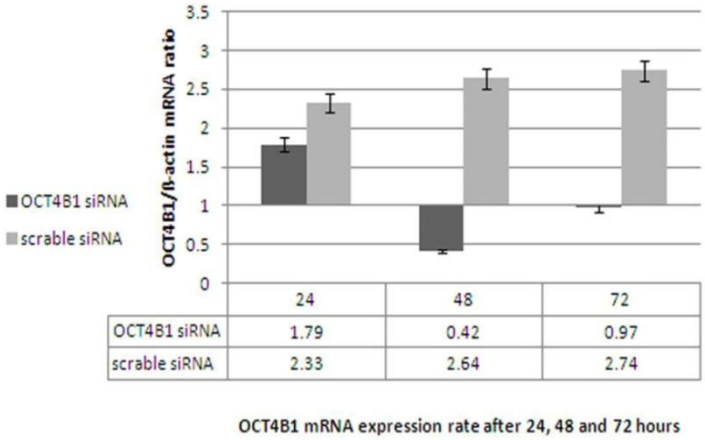

Figure1. Demonstrates OCT4B1 expression after 24, 48 and 72 hours of siRNA transfection in AGS cell line. Y axis showed rate of OCT4B1 variance expression compared to ß-actin (housekeeping gene as control) and X axis showed AGS cell line transfected with OCT4B1 siRNA (test) and scramble siRNA transfected cell (control) after 24, 48 and 72 hours of transfection.

Elevated apoptosis and decreased viability of studied tumor cell lines treated with specific OCT4B1 siRNA

Flow cytometry analysis of the cells stained with Annexin V and PI demonstrated that two days after siRNA transfection up to 34% of the test group cells

(transfected by specific OCT4B1 siRNA) underwent apoptosis, in comparision to control cells (transfected by scrambled sequence siRNA)(Figure not shown). MTT assay also indicated reduced cell activity after OCT4B1 siRNA transfection compared to cells

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[117]

transfected by scrambled sequence siRNA and mock transfected control cells. (Figure 2).

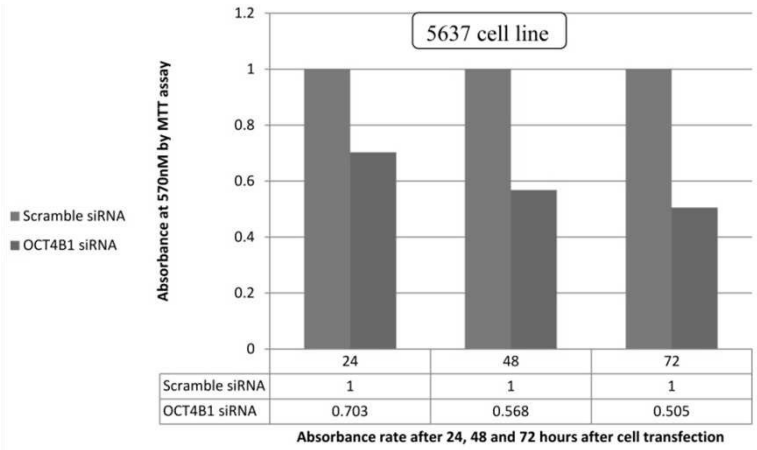

Figure 2. The figure shows OCT4B1 siRNA transfected cells had lower viability in compare to control groups (scramble siRNA transfected cells).

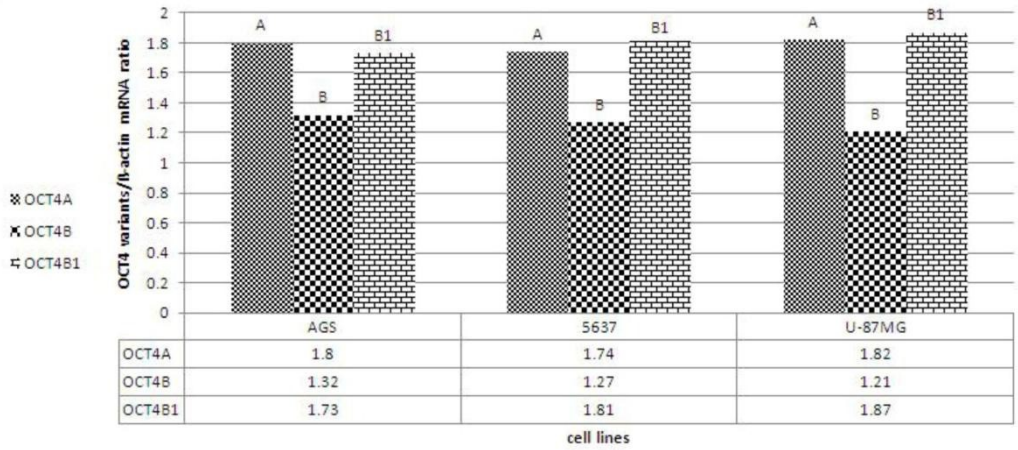

Figure 3. Demonstrates level of OCT4 variants expression in studied tumor cell lines. Y axis shows OCT4B1 variance mRNA expression level compared to ß-actin (as housekeeping control gens) and X axis indicates three studied tumor cell lines (AGS, 5637 and U87MG).

Table 3. TNF ligand genes family up/down regulated after OCT4B1 suppression in 5637, AGS and U87MG tumor cell lines.

Gene bank number		Gene symbol	Protein/gene function	**Cell lines/ Fold changes**		
				5637	**AGS**	**U87MG**
NM-000074	**TNFSF5**	Tumor necrosis	**1.67**	**1.51**		**1.46**
NM-000639	**TNFSF6**	Tumor necrosis	**-3.51**	**2.29**		**-1.49**
NM-000595	**TNFSF1**	Tumor necrosis	**-3.53**	**-6.31**		**-1.7**
NM-000594	**TNF**	Tumor necrosis	**13.95**	**5.32**		**2.3**
NM-003810	**TNFSF10**	Tumor necrosis	**-12.28**	**-4.26**		**-10.54**
NM-001252	**TNFSF7**	Tumor necrosis	**-10.18**	**-6.68**		**-6.18**
NM-001244	**TNFSF8**	Tumor necrosis	**-8.21**		**-12.64**	**-7**

Quantitative Real-time PCR results

Our data revealed that all three variants were expressed

TNFSF6 were down-regulated between 2-10 fold (Table 3 and Figure 4). In the TNF receptor transcript family, six transcripts including, TNFRSF10A, TNF transcript was up-regulated (>8 fold), while TNFSF8, TNFSF7, TNFSF10, TNFSF1 and

in the studied cell of the studied transcripts in three cell lines is almost similar (Table 3 and 4). In TNF ligand transcript family,

TNFRSF10B, TNFRSF11B, TNFRSF1A, TNFRSF21 **and TNFRSF25 were up-regulated by more tahn 5 fold,** while TNFRSF9 and CD27 were down-regulated by more than 3 fold (Table 4 and Figure 5).

[118]

Table 4. TNF receptor genes family up/down regulated after OCT4B1 suppression in 5637, AGS and U87MGtumor cell lines.

Gene bank number	Gene Symbol	Protein/gene function	Cell Lines/Fold changes 5637	AGS	U87MG
NM-001250	**TNFRSF5**	Tumor necrosis	**1.8**	**-3.59**	**-1.35**
NM-000043	**TNFRSF6**	Tumor necrosis	**-2.83**	**-2.09**	**-1.24**
NM-002342	**TNFRSF3**	Tumor necrosis	**1.63**	**-2.97**	**3.16**
NM-003844	**TNFRSF10A**	Tumor necrosis member 10a	**6.74**	**9.72**	**7.53**
NM-003842	**TNFRSF10B**	Tumor necrosis member 10b	**6.42**	**9.65**	**12.48**
NM-002546	**TNFRSF11B**	Tumor necrosis member 11b	**12.4**	**9.26**	**8.89**
NM-001065	**TNFRSF1A**	Tumor necrosis member 1a	**17.29**	**-**	**9.14**
NM-014452	**TNFRSF21**	Tumor necrosis member 21	**12.4**	**12.91**	**11.1**
NM-003790	**TNFRSF25**	Tumor necrosis member 25	**5.83**	**8.94**	**8.01**
NM-001242	**CD27**	CD27 molecule	**-3.8**	**-9.91**	**-4.85**
NM-001561	**TNFRSF9**	Tumor necrosis member 9	**-8.35**	**-12.04**	**-7.68**

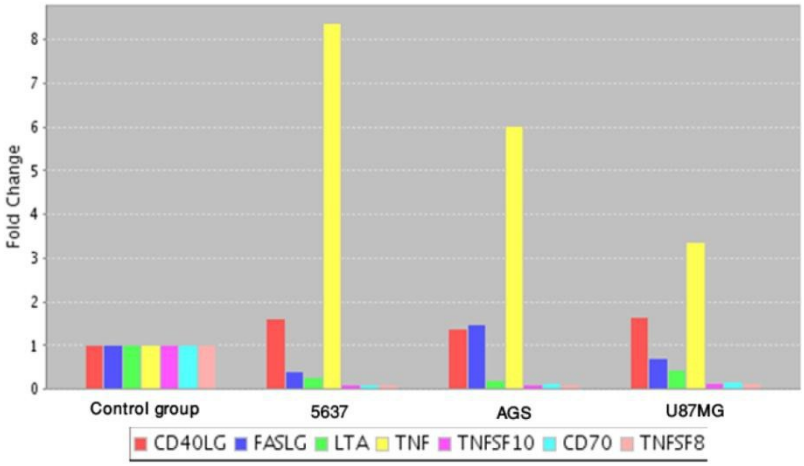

Figure 4. Demonstrates TNF ligand family gene expression 48 hours after siRNA transfection in studid tumor cell lines. Y axis showed fold gene regulation and X axis showed 25 genes of anti-apoptotic gene family in test and control groups. group1; AGS, group2; 5637 and grou3; U87MG tumor cell lines.

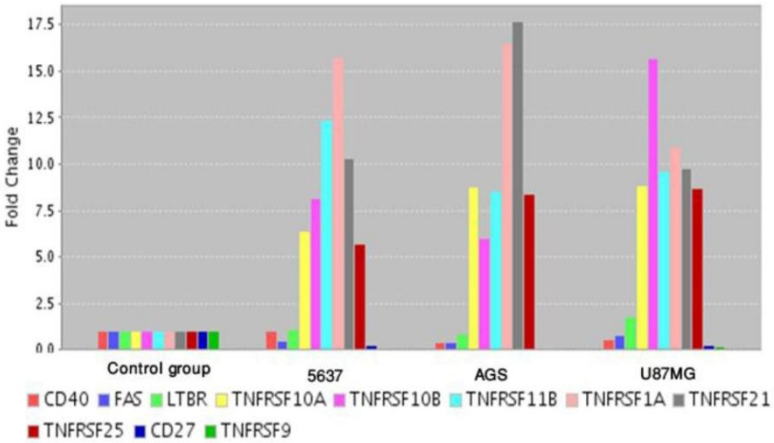

Figure 5. Demonstrates TNF receptor family gene expression 48 hours after siRNA transfection in studid tumor cell lines. Y axis showed fold gene regulation and X axis showed 25 genes of anti-apoptotic gene family in test and control groups. group1; AGS, group2; 5637 and grou3; U87MG tumor cell lines.

Discussion

In general, development of tumor is characterized by increasing of tumor cell mass. This could be due to increased proliferation and/or decreased apoptosis, or both. A coordination and balance between cell proliferation and apoptosis is crucial for normal development. [15]

TNF ligand and receptors are two gene families with a critical role in apoptosis pathway, they alter TNF ligands and receptors expression, and dramatically change apoptosis rates.[16]

There are contradictory reports for expression of OCT4 in cancer cells and cancer tissues, some are impressed to express this gene in cancer tissues, while other reports indicate OCT4 does not express in cancer cell line and tissues.[17,18] However, it has been documented that OCT4 expression induces cell stemness and down-regulation or silencing caused differentiation.[19] Atlasi et al. defined and introduced a novel variant of OCT4, designated as OCT4B1, which is highly expressed in tumor cell lines

Most TNF ligands and receptors are expressed in the and down-regulated upon differentiation.[1] Consistant with our results, Asadi and collaborators revealed that OCT4B1 has anti-apoptotic potency.[5] Farashahi et al. also determined a direct relation between stress (heat shock) and up-regulation of OCT4B1.[20] Over-expression of OCT4B1 in tumor cell lines and cancer tissues in compared to normal cells and tissues and so, anti-apoptotic potency of this variant, suggests OCT4B1 play an important role in cancer origination and progression in tumor cell lines.

In the current research all three OCT4 variants were expressed in studied cell lines and following the suppression of the OCT4B1, the mRNA rate of this variant was significantly decreased compared to cells that transfected by scrambled siRNA. OCT4A expression maybe related to OCT4 pseudogenes that have the most similarity to OCT4A variant.[21] OCT4B is expressed in a variety of cells (somatic and tumor

immune system although several studies have emphasized expression of these ligands and receptors in cancer cell lines with different patterns.[22,23]

Due to the fact that TNF Superfamily molecules induce apoptosis via extrinsic pathway, based on our results it may be concluded that OCT4B1 suppression leads to apoptosis via extrinsic pathway.

Previous studies showed, siRNA transfections may caused different behavior in transfected cells especially in different cell lines,[24] but our results revealed that expression pattern of studied gene families in three studied tumor cell lines was the same, hence it may be concluded that OCT4B1 plays an alike role in tumor cell lines and inhibits apoptosis, maybe, via down-regulation of TNF superfamily molecules.

To the best of our knowledge this is the first investigation regarding the relation between OCT4B1 suppression and gene expression profile of TNF ligands and receptors in tumor cell lines. As the results have shown, there is a tight collaboration between suppression of this variant and up/down regulation of the studied TNF ligands and receptors genes. Therefore, the data demonstrated that OCT4B1 potentially targets TNF ligands and receptors members, as changes the self-renewal genes.[25]

In conclusion, the present study indicates that OCT4B1suppression triggered the cells toward apoptosis and reduced cell viability, so, suppression of this variant has significant effects on expressional pattern of TNF ligand and receptors genes family. Based this findings, OCT4B1, perhaps plays a critical role in Commutation of normal cell to cancer situation so, it may be considered in later investigation around cancer.

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cell lines and tissues) and studies have revealed this variant does not have a role in stemness state. Results of Annexin V-FLOS test showed that approximately **29% of OCT4B1 siRNA transfected cells unedrwent** apoptosis in compared to scrambled siRNA transfected cells. Based on these results it can be concluded that following suppression of OCT4B1, the apoptosis

References

1. Atlasi Y, Mowla SJ, Ziaee SA, Gokhale PJ, Andrews PW., OCT4 spliced variants are

differentially expressed in human pluripotent and nonpluripotent cells. *Stem Cells* 2008;26:3068-3074.

pathway is activated and triggers further processes of **apoptosis. Atfer OCT4B1 suppression (48 hours),** mRNA synthesis rate of the five studied TNF ligand

genes including: TNFSF8, TNFSF7, TNFSF10, TNFSF1 and TNFSF6 were decreased (down-regulated) by 2-10 fold and expression of TNF gene was elevated (up-regulated) more than 8 fold (Table 2).

From 11 TNF receptor genes analyzed, 7 genes including: TNFRSF10A, TNFRSF10B, TNFRSF11B,

2. Tai MH, Chang CC, Kiupel M, Webster JD, Olson LK, Trosko JE., Oct4 expression in adult human stem cells: evidence in support of the stem cell theory of carcinogenesis. *Carcinogenesis* 2005;26:495-502.

TNFRSF1A, TNFRSF21, TNFRSF25 and TNFRSF9 were up-regulated by (5-15 fold), and CD27 and FAS genes were down-regulated 2-3 fold, and TNFRSF3 (LTBR) gene expression shows no change (Tables 3).

3. Kokuryo T,Yokoyama Y,Nagino M., Recent advances in cancer stem cell research for cholangiocarcinoma. *J Hepatobiliary Pancreat Sci* 2012;19:606-613.

4. Podberezin, M., Wen J., and Chang C., Cancer Stem Cells: A Review of Potential Clinical Applications. *Arch Pathol Lab Med* 2012.

5. Asadi MH, Mowla SJ, Fathi F, Aleyasin A, Asadzadeh J, Atlasi Y., OCT4B1, a novel spliced variant of OCT4, is highly expressed in gastric

17. Cantz T, Key G, Bleidissel M, Gentile L, Han DW, Brenne A, Schöler HR., Absence of OCT4 expression in somatic tumor cell lines. *Stem Cells* 2008;26:692-697.

cancer and acts as an antiapoptotic factor. *Int J Cancer* 2011;128:2645-2652.

6. Cotter, T.G., Apoptosis and cancer: the genesis of a

18. Berg, J.S. and Goodell M., An argument against a role for Oct4 in somatic stem cells. *Cell Stem Cell* 2007;1:359-360.
7. Green, D., Apoptotic pathways: paper wraps stone blunts scissors. *Cell* 2000;102:1-4.
8. Aggarwal BB, Shishodia S, Ashikawa K, Bharti AC., The role of TNF and its family members in inflammation and cancer: lessons from gene deletion. *Curr Drug Targets Inflamm Allergy* 2002;1:327-341.
9. Vinay, D.S. and Kwon B., TNF superfamily: costimulation and clinical applications. *Cell Biol Int* 2009;33:453-465.
10. Aggarwal, B., Signalling pathways of the TNF research field. *Nat Rev Cancer* 2009;9:501-507.

19. Wang, X. and Dai J., Concise review: isoforms of OCT4 contribute to the confusing diversity in stem cell biology. *Stem Cells* 2010;28:885-893.
20. Farashahi Yazd E, Rafiee MR, Soleimani M, Tavallaei M, Salmani MK, Mowla SJ., OCT4B1, a novel spliced variant of OCT4, generates a stable truncated protein with a potential role in stress response. *Cancer Lett* 2011; 309: 170-175.
21. Suo G, Han J, Wang X, Zhang J, Zhao Y, Zhao Y, Dai J., Oct4 pseudogenes are transcribed in cancers. *Biochem Biophys Res Commun* 2005;337:1047-1051.

superfamily: a double-edged sword. *Nat Rev Immunol* 2003;3:745-756.

11. Ware, C., The TNF superfamily. *Cytokine Growth Factor Rev* 2003;14:181-184.

22. Collette Y, Gilles A, Pontarotti P, Olive D., A co-evolution perspective of the TNFSF and TNFRSF families in the immune system. *Trends Immunol* 2003;24:387-394.

12. Bernas, T. and Dobrucki, J., Mitochondrial and nonmitochondrial reduction of MTT: interaction of MTT with TMRE, JC-1, and NAO mitochondrial fluorescent probes. *Cytometry* 2002;47:236-242.

13. Demchenko, A., The change of cellular membranes on apoptosis: fluorescence detection. *Exp Oncol* 2012;34:263-268.

23. Pelekanou V, Notas G, Theodoropoulou K, Kampa M, Takos D, Alexaki VI, Radojicic J, Sofras F, Tsapis A, Stathopoulos EN, Castanas E., Detection of the TNFSF members BAFF, APRIL, TWEAK and their receptors in normal kidney and renal cell carcinomas. *Anal Cell Pathol (Amst)* 2011;34:49-60.

14. Koopman G, Reutelingsperger CP, Kuijten GA, Keehnen RM, Pals ST, van Oers MH., Annexin V for flow cytometric detection of phosphatidylserine

24. Endo-Takahashi Y, Negishi Y, Kato Y, Suzuki R, Maruyama K, Aramaki Y., Efficient siRNA expression delivery using novel siRNA-loaded Bubble 1994;84:141 liposomes and ultrasound. *Int J Pharm*

15. Guo, N. and Peng Z., MG132, a proteasome inhibitor, induces apoptosis in tumor cells. *Asia Pac J Clin Oncol* 2012.
16. Neznanov N, Kondratova A, Chumakov KM, Angres B, Zhumabayeva B, Agol VI, Gudkov AV., Poliovirus Protein 3A Inhibits Tumor Necrosis Factor (TNF)-Induced Apoptosis by Eliminating the TNF Receptor from the Cell Surface. *J Virol* 2001;75:10409-10420.

2012;422:504-509.
25. Afsah Hejri S. J, Mirzaei M. R, Kazemi M., Hassanshahi Gh. and Mahmoodi M., OCT4B1 Down-Regulates Self-Renewal Genes in Cancer Cell Lines. *Pharm Sci* 2014;19:117-123.